

Theory is in the Eye of the Beholder: Exploring Difficulties with Validating Intervention Mechanisms

Theory is in the Eye of the Beholder

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ABSTRACT

Behavior change researchers frequently base their interventions on theory, targeting specific mechanisms of change to help users achieve their goals. Thus HCI researchers have sought to examine whether their intervention impacts the intended mechanism of change, in addition to evaluating the overall effect of the intervention. Yet an open question remains: how do we know our interventions successfully target the mechanisms we intend? We present results from two validation studies and one user study showing the difficulties of ensuring that behavior change interventions target the mechanisms designers intend. Our findings indicate that experts disagree about what mechanism an intervention targets, that expert consensus on this matter can be hard to achieve, and that end users' reflections indicate they may follow different mechanisms than those predicted by experts. We recommend that researchers collect data about multiple potential mechanisms that their intervention could operate through, rather than the common single-mechanism approach.

CCS CONCEPTS

• Human-Centered Computing; • Human-Computer Interaction; • HCI Design and Evaluation Methods;

KEYWORDS

health behavior change research, health behavior change theories, intervention validation, intervention design methods

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1 INTRODUCTION

Much HCI research has been devoted to supporting health behavior change, including behaviors such as increasing physical activity (e.g. [9, 13, 17, 19]) and smoking cessation [6]. These interventions are often based on theories of behavior and behavior change developed by researchers in behavioral science and psychology. Common models include the Health Belief Model [7], the Trans-Theoretical Model [23], the Theory of Planned Behavior [3], the COM-B model [21], and the Integrated Behavioral Model [22]. These models show how different *constructs*, or the “fundamental components or ‘building blocks’ of behavioral theory” [11], relate to each other to determine behavioral intention or behavior. A *mediator* or *mechanism* refers to how an intervention works [11, 18], for example what constructs an intervention targets to effect behavior change. To help design these interventions appropriately, researchers have developed protocols for intervention designers to choose an appropriate theory or theories to apply, and on which to base their intervention and evaluation plan [5, 16].

However, behavior change is complex and difficult, and interventions can fail for a variety of reasons, ranging from internal problems with their design to external obstacles that are separate from the intervention itself. For this reason, researchers have called for evaluations of behavior change interventions based not just on whether the intervention succeeds, but on whether the intervention actually affects the mechanism of change it targets [11, 12, 14, 15]. For example, a behavior change intervention to build self-efficacy for exercise can be evaluated not just on whether users exercise more, but also on whether their self-efficacy for exercise has improved by using the intervention.

Such calls have typically emphasized the intervention's *intended* mechanism of change. But how should we validate that the intervention is actually targeting the *intended* mechanism, and not another mechanism of change? Although researchers have tried to map behavior change intervention techniques [1, 21] onto theoretical mechanisms of change via expert consensus [8], prior work

suggests some variation in experts' interpretation of how behavior change interventions are supposed to work, and few have examined whether end users respond as expected to behavior change interventions targeting specific mechanisms.

We present insights from a three-part study to create and evaluate early prototypes for one particular context—encouraging patients to speak up about concerns in the hospital. Encouraging patients to speak up has been the subject of a variety of interventions [24], but it has rarely been examined through the lens of a theoretical model of behavior. We approached speaking up as a behavior to explore theory-based patient-facing interventions that should encourage patients to voice their concerns to clinicians. Here, we discuss findings from our design process showing the difficulty of validating interventions via expert consensus, and illustrating how even when expert consensus is achieved, it may not line up with user responses to interventions. We conclude with a discussion of the implications of these findings for behavior change research in HCI and suggestions for future research.

2 RELATED WORK: THE INTEGRATED BEHAVIORAL MODEL

Behavioral scientists have come up with many behavior change models, some of which depict behavior as influenced by a variety of constructs. In this section we describe one such model, the Integrated Behavioral Model (IBM) [22], on which this research is based. The IBM builds upon the better-known Theory of Reasoned Action [10] and Theory of Planned Behavior [3], but offers more granular constructs. The model posits intention to perform a behavior as one of the central factors for whether someone performs the behavior (along with other factors such as environmental constraints). The IBM has six mediators that contribute to behavioral intention [22]. *Experiential attitude* is defined as someone's affective feelings about a behavior (e.g., is it pleasant or unpleasant); *instrumental attitude* is someone's cognitive attitude towards the behavior (e.g., is it a good idea or bad idea, what kinds of outcomes will result). *Injunctive norms* are a person's beliefs about whether other people would approve of the behavior or think that the person should do the behavior. *Descriptive norms* are a person's beliefs about whether other people perform the behavior or not. *Self-efficacy* is a person's confidence in their ability to carry out the behavior. Finally, *perceived control* is the degree to which a person believes carrying out the behavior is under their control at all. These mediators are potential mechanisms of behavior change that can be targeted by interventions.

The IBM is just one behavior change model out of many, and has its fair share of weaknesses. For example, Hekler et al. state that many behavioral scientists consider self-efficacy and perceived control to be essentially the same thing [11] because they both relate to a person's perception of their ability to perform a behavior [2], even though they are distinct from each other in the IBM. However, this very granularity makes the IBM a good model for offering researchers a broad variety of potential mechanisms of change on which to base interventions, as was done in this work. The granularity of constructs also allows for greater precision in evaluating how expert assessments line up with user responses to an intervention.

3 METHODS

We summarize the methods here of three studies: a survey of behavior-change experts, interviews with behavior change experts, and interviews with patients and caregivers. All study procedures were approved by the researchers' Institutional Review Board.

3.1 Study 1 Methods: Expert Survey

As mentioned earlier, this work was done in the context of developing interventions to encourage hospitalized patients to speak up about issues in their care, based on the six mechanisms of change in the IBM, described above. To achieve this goal, we created low-fidelity prototypes of interventions, and a survey for experts to map the prototypes to the mechanism of change they thought each prototype operated through (see Fig. 1A for example survey question). Our initial plan was to validate each prototype through expert consensus on the mechanism of the prototype. Experts were defined as people who had an advanced degree (i.e., higher than a bachelor's degree) in a field pertaining to HCI or behavioral science, and who demonstrated knowledge of the IBM through an in-survey assessment. Consensus was originally defined as an average Cohen's Kappa indicating at least moderate agreement, although that approach was changed to a simple majority (described in Section 4.1). The survey was piloted with graduate students in fields related to HCI and then conducted in two rounds, with three expert respondents in Round 1 and four in Round 2. In total 24 intervention prototypes were presented in the survey, with the expectation that only some of these prototypes would achieve expert consensus and be used in interviews with patients.

Experts were recruited through a combination of personal and professional networks, via word of mouth. Respondents ($n=7$) included 3 post-docs, 3 faculty, and 1 industry researcher across the disciplines of information science, health behavior or behavioral medicine, computer science, and nutrition. Unless respondents indicated prior experience with the IBM ($n=1$), respondents were given an overview of the model with definitions of constructs including the six mechanisms of change described earlier, and then quizzed about their understanding of each construct. All respondents passed with 92% accuracy (1 respondent) or 100% (5 respondents).

3.2 Study 2 Methods: Expert Interviews

Several expert respondents commented that they had found the survey challenging to complete. Because of these comments, we asked expert respondents to participate in brief, semi-structured interviews about their experience taking the survey. Our goal was to investigate our validation method and make sure our results were reliable, as well as to understand the reasons for the comments. Three expert respondents (from Study 1) and one graduate student who had piloted the survey (but was not officially a participant in Study 1) consented to participate. In findings reported below, these participants are referred to with the prefix "E" for "Expert" (e.g., "E01").

3.3 Study 3 Methods: Patient and Caregiver Interviews

As described earlier, the primary goal of our study was to explore ways to motivate patients to speak up about issues related to their hospital care. Through the survey (Study 1), we refined the total



Figure 1: Prototypes used in studies. (A) Example survey question from Study 1. (B) Graph showing that 95% of clinicians think patients should speak up about their concerns. (C) Bar graph showing how often patients speak up. (D) Bar graph of errors patients have prevented by speaking up.

set of interventions to 12 prototypes. We used these prototypes in 40-60 minute semi-structured interviews with hospitalized patients and caregivers at two hospital sites: Seattle Children's Hospital and Virginia Mason Hospital. From Seattle Children's we recruited 10 patient participants. 6 caregivers of patients also chose to participate in the interview. Patient participants ranged in age from 13-21 (mean 16.1, median 16), and caregiver participants were aged 37, 55, and the rest not collected. From Virginia Mason we recruited 15 patient participants, ranging in age from 26 to 79 (mean 56.8, median 57). In addition, one caregiver (the participant's spouse) decided to participate in the interview. Over both sites, 17 participants identified as female and 11 as male. 17 participants identified as White, 2 as Latinx, 3 as Asian, 2 as Black, 1 as White and Asian, 1 as White and Native Hawaiian, and 1 as Asian and Latinx. Participants are identified with A## (Virginia Mason) or Y## (Seattle Children's), with caregiver participants identified with an appended -C (e.g., A05, A05-C). Participants were drawn from multiple medical services at both sites. The interviews were transcribed and analyzed by two members of the research team with a deductive codebook derived from IBM constructs.

4 FINDINGS

We present findings below, by study. We found low consensus among expert respondents on our prototypes in the survey. Our interview findings showed that although lack of familiarity with the model posed a problem, more difficulties arose from the complexity of the design and experts' assumptions about the interrelation of mechanisms of change, which were outside of the model. Patients shared this type of assumption. Moreover, we found to our surprise that patients and caregivers sometimes responded to prototypes differently from what experts predicted.

4.1 Study 1 Findings: Low Consensus Among Experts

Our original intention was to validate prototypes as acting through particular mechanisms by taking average values of Cohen's Kappas and using moderate agreement to define "expert consensus" on the mechanism of a prototype. However, achieving even moderate agreement on prototypes proved difficult. Out of the 17 prototypes presented in the first round validation survey, only 6—less than

half—achieved an average kappa value of .41 or more (moderate agreement according to Cohen [20]). In the second round validation survey, none of the 7 prototypes achieved an average kappa value of .41 or more. Moreover, prototypes did not always validate for the mechanism for which they had been intended. For instance, one prototype had been intended to target the *instrumental attitudes* mechanism by tracking the efficacy of speaking up in preventing medical errors (see Fig. 1D) – in other words, by shaping outcome expectations. Instead, expert respondents unanimously perceived intervention as targeting the *descriptive norms* mechanism, by showing that patients *were* speaking up.

4.2 Study 2 Findings: Reasons for Low Consensus

Low consensus among experts was due to many factors, including lack of knowledge about the model and complexity of intervention prototype design. Interestingly, experts struggled to map prototypes onto specific mechanisms of change because they felt mechanisms were inherently interrelated in a way not represented by the model.

4.2.1 Knowledge of the Model. As described earlier, experts demonstrated knowledge of the IBM on the in-survey assessment. Expert interviews revealed that some expert respondents encountered minor difficulties with model. For instance, E02 explained, “*the more familiar I was with the concept the more . . . comfortable I was applying it. But for some of these newer concepts to me . . . I had a little bit more trouble mapping those onto [interventions].*” However, discomfort with the model was not necessarily a total roadblock for expert respondents. E01 described flipping back and forth between survey pages to double check construct definitions when mapping interventions to make sure she was remembering them accurately. In addition, not all experts felt discomfort with the model. Overall, experts’ comments during interviews revealed that their difficulty responding mapping interventions to mechanisms of change sprang in large part not from unfamiliarity with the IBM but from the difficulty of mapping intervention prototypes onto individual mechanisms, described next.

4.2.2 Complex Prototype Design. When a prototype was complex, experts found it more challenging to map it to mechanisms, because different parts of the design could be seen as relating to different mechanisms, even if the intervention as a whole had been designed for a specific mechanism. For instance, one prototype shown in the survey Round 2 had been designed with three parts, each based on one of Bandura’s proposed methods for building *self-efficacy* [4]: a prompt for patients to reflect on past times they’d had difficult conversations with authority figures, a prompt for patients to imagine how the conversation might go, and a prompt for patients to look at fictional example conversations to help them prepare for speaking up to their clinicians (see Fig. 1A). E01 and E02, who were both very knowledgeable about self-efficacy and its application to interventions in the context of behavior change, saw this *self-efficacy* intervention as extending beyond *self-efficacy*. As E01 put it, “*I mean for example this one has like three things. . . like ‘see some examples of how other people can speak up about their concerns’ . . . I know that that’s. . . the descriptive norm? But the other two are, ‘imagine how the conversation might go,’ ‘reflect on your skills’ . . .*

those are about you doing some form of reflection . . . And so . . . the reason why it’s more complicated, this one, is because it was three things . . .”

4.2.3 Assumptions About the Interrelation of Mechanisms of Change.

Despite their complaints about the complexity of the prototypes, even the simplest prototypes could be difficult for experts to map to one mechanism, because experts held implicit assumptions about connections between mechanisms that are not represented in the IBM. For instance, one prototype (see Fig. 1B) consisted of a simple pie graph showing poll results, stating that 95% of clinicians think it is important for patients to speak up about issues in their care. The prototype was intended to target *injunctive norms* (beliefs about whether others would approve of what you do), and by simple majority this prototype passed validation only for *injunctive norms*. However, E04 had individually mapped the prototype to both *injunctive* and *descriptive norms*. In the interview, E04 explained that 95% was such a high number that if 95% of clinicians felt a certain way, it also suggested something about what patients and clinicians were actually doing (*descriptive norms*). In other words, E04 felt that someone could not hold this belief about *injunctive norms* without also believing something about *descriptive norms*—the two were not independent of each other, as they are supposed to be in the model.

E03 similarly found connections between how mechanisms operated. For instance, one prototype offered a bar graph showing how often patients speak up about their concerns in the hospital (see Fig. 1C). E03 mapped this prototype to both “*descriptive norm[s] as a first instance and maybe self-efficacy—if some patients see other patient[s] that are used to tell . . . clinicians about their problems, maybe they can feel more confident [to speak up] . . .*” Thus E03 mapped a prototype to two mechanisms because of implicit assumptions about a causal connection between the two, even though those two mechanisms are not causally connected in the IBM (or many other behavioral models).

4.3 Study 3 Findings: Expert vs. Patient Assessments

Just like experts, interviews with patients also revealed their assumptions about the interrelations of mechanisms. A01, looking at the same prototype as E03 (Fig. 1C), understood the prototype as saying something about how often other patients spoke up (*descriptive norms*), but also said the graph made her feel “*like people have your back*” and that other patients would approve of her for speaking up (*injunctive norms*).

Moreover, we found that patients sometimes differed from experts in how they responded to prototypes. For instance, experts agreed unanimously that one prototype—a bar graph tracking how many errors patients had prevented by speaking up (Fig. 1D)—acted only through *descriptive norms*. Yet one caregiver, Y01-C, was enthused about this prototype, not because it illustrated that patients were speaking up (*descriptive norms*), but because it could influence expectations about the outcomes of speaking up (*instrumental attitudes*). Another patient, A06, likewise praised this prototype because it illustrated of the fact that “*people actually hear you*” and “*your speaking up makes a difference*”—once again suggesting that

this intervention could function through instrumental attitudes rather than descriptive norms.

5 DISCUSSION

Our findings yield implications for how to evaluate behavior change interventions, as well as implications for future work that must be done on the Integrated Behavioral Model. We describe both below.

5.1 Implications for Evaluating Behavior Change Interventions

Researchers have called for increased attention to how behavior change interventions impact specific mechanisms of change [11, 12, 14, 15]. Here we present unexpected findings from an attempt to map prototype interventions to specific mechanisms of change, highlighting some of the difficulties inherent in this mapping. Which mechanism should be evaluated: the mechanism that the designer of the behavior change intervention intends to target? The mechanism that other experts believe the intervention operates through? Or the mechanism that data from end users suggest may actually be in effect? Our findings showed that these mechanisms may not be the same.

Our study suggests that researchers should measure all the mechanisms of change on which an intervention is likely to function. Thus far, calls for directly measuring the impact of behavior change interventions on mechanisms of change [11, 15] have been concerned with the risk of missing a real effect if an intervention fails, and the risk of not accurately testing the hypothesis in the first place. Thus, these calls have focused on intended mechanisms. Our findings suggest that intended mechanisms may not capture the whole truth of an intervention—that interventions work in unexpected ways. Interventions with complex designs may unintentionally leverage more than one mechanism. In addition, individuals may respond to interventions in unexpected ways, possibly because they make implicit assumptions not captured by the behavior change models upon which researchers base their designs.

This issue is important for two reasons. First, if a behavior change intervention fails, it might still have a real effect via an unexpected mechanism, which researchers might fail to capture. Second, if the intervention works, it might work via an unintended mechanism rather than an intended mechanism, or even via more than one mechanism. In both cases, the implication of our findings is that researchers should plan to collect data about multiple potential mechanisms that their intervention could operate through. Although capturing the entire universe of potential mechanisms is not feasible, we suggest following recommendations from Hekler et al. [11] and using the *entirety* of the model the intervention is based on as a guide – in other words, evaluating whether the intervention is working by any of the mechanisms in the model, rather than only the intended mechanism. This evaluation could be approached qualitatively, as was done here, or using validated questionnaires where available.

5.2 Implications for the Integrated Behavioral Model

In our findings, both experts and users (patients) held implicit assumptions about relationships between mechanisms of change that

were not represented in the IBM. Whether such connections always exist in the real world—for example, whether *injunctive norms* actually suggest something about *descriptive norms*—our findings illustrate that people’s beliefs about these mechanisms are not always independent. A better understanding of the connections people draw between the mechanisms in the IBM, and other behavioral models, could drive development of new behavior change interventions. For example, a finding that beliefs about descriptive norms affect self-efficacy could suggest that an intervention to promote exercise could build self-efficacy in the face of common obstacles like weather by highlighting, for instance, how many other people are also exercising even if the weather is bad. Our findings further underscore the importance of understanding the implicit assumptions end-users, rather than experts, hold about the relationships between mechanisms of change in different health contexts, to shape intervention design.

6 LIMITATIONS AND FUTURE WORK

There are several limitations on this work. The sample of experts was quite small. In addition, because only one model was studied, it is difficult to say the degree to which our results were the result of weaknesses with the IBM (e.g., poor delineation of constructs), as opposed to difficulties inherent to mapping interventions onto particular mechanisms of change. Another potential contributing factor, which could not be explored in this study given the small sample size, is the background of expert participants. For instance, researchers trained in behavioral science may have a different perspective from researchers trained in HCI. More work is needed to fully explore the difficulties inherent in validating prototypes for behavior change.

7 CONCLUSION

Calls have grown for researchers to measure not just the overall effect of their behavior change interventions, but also whether the intervention impacts its intended mechanism of change. However, our results show that behavior change interventions can impact mechanisms of change that are different from what is intended, and that end users may not respond to interventions as expected. These findings highlight the importance of considering alternate mechanisms that behavior change interventions could potentially impact, to deepen our understanding of how interventions succeed or fail.

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